

TRUE Simulation Experiment Report

P0/P1 Refactored Round

Trust-Reliability Unified Engineering
Multi-Agent Collaboration Simulation

100 runs x 200 rounds x 4 scenarios x 6 groups
Fixed task-flow pairing | Bootstrap CIs | Bonferroni correction

Executive Summary

This report presents the P0/P1 refactored experimental results for the TRUE (Trust-Reliability Unified Engineering) multi-agent collaboration framework. All groups within each Monte Carlo run face the identical task sequence, ensuring that outcome differences are attributable to selection mechanisms rather than task luck.

Key changes: (1) Fixed task-flow pairing; (2) Removed group-specific observation manipulation for A8; (3) Renamed MOO to TTB (Trust-Targeted Baseline); (4) Added ablation variants TRUE-C, TRUE-E, TRUE-N; (5) Wilcoxon signed-rank test + bootstrap 95% CIs + Bonferroni correction.

Scenario Definitions

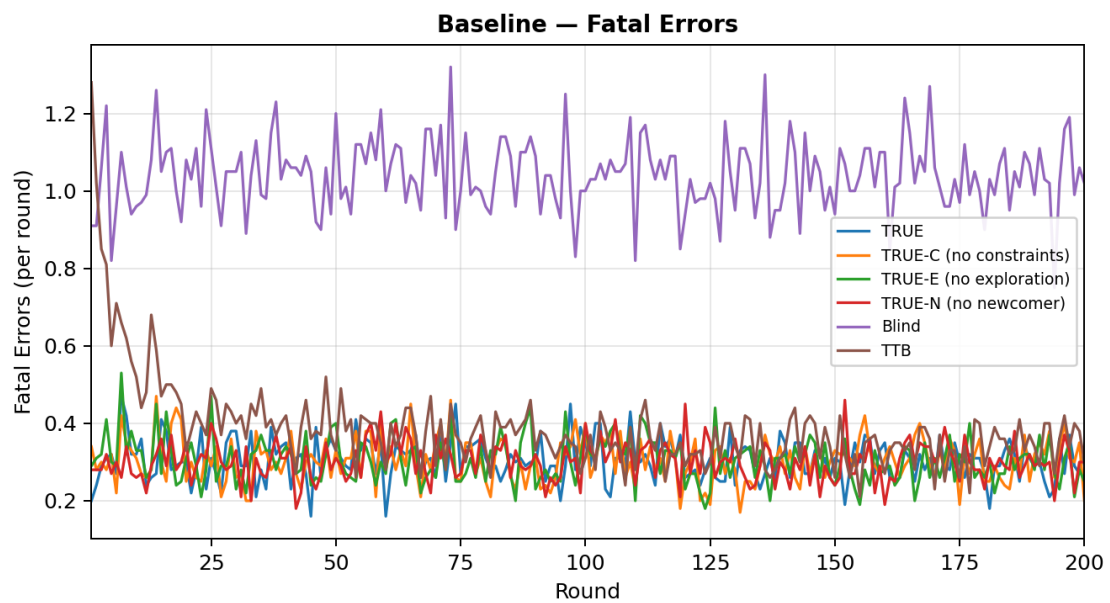
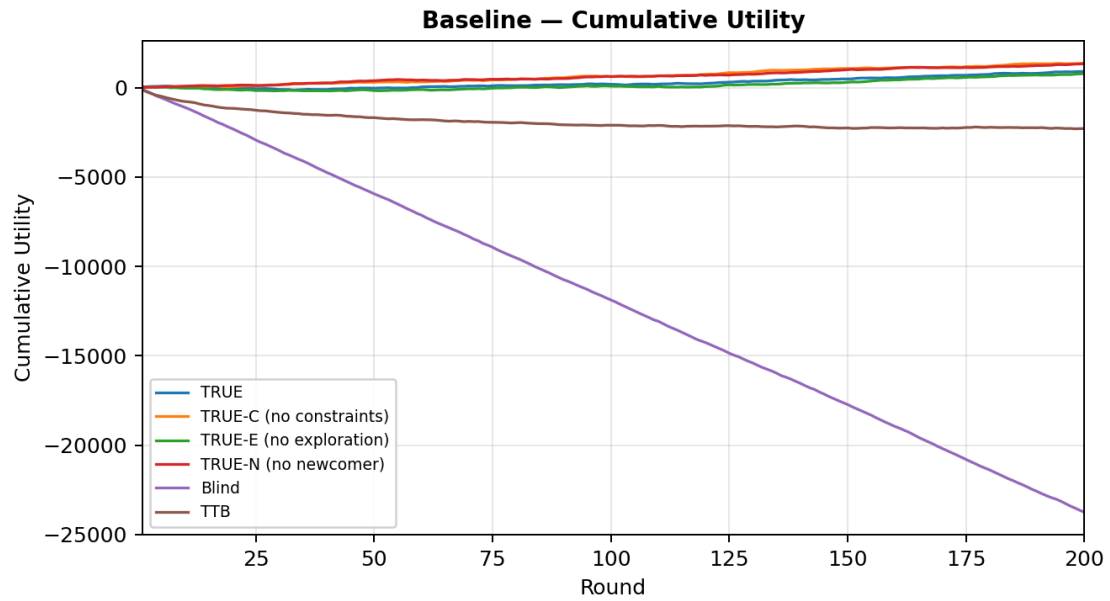
Scenario	Description
Baseline	Standard engineering collaboration with moderate observation noise.
Safety-Critical	High-stakes safety environment with harsher penalties and stronger error propagation.
Observation-Manipulated	Observation-contaminated environment where surface quality diverges from true quality.
Utility-Trust-Misalignment	Local phases where trusted incumbents underperform and low-history entities hold latent value.

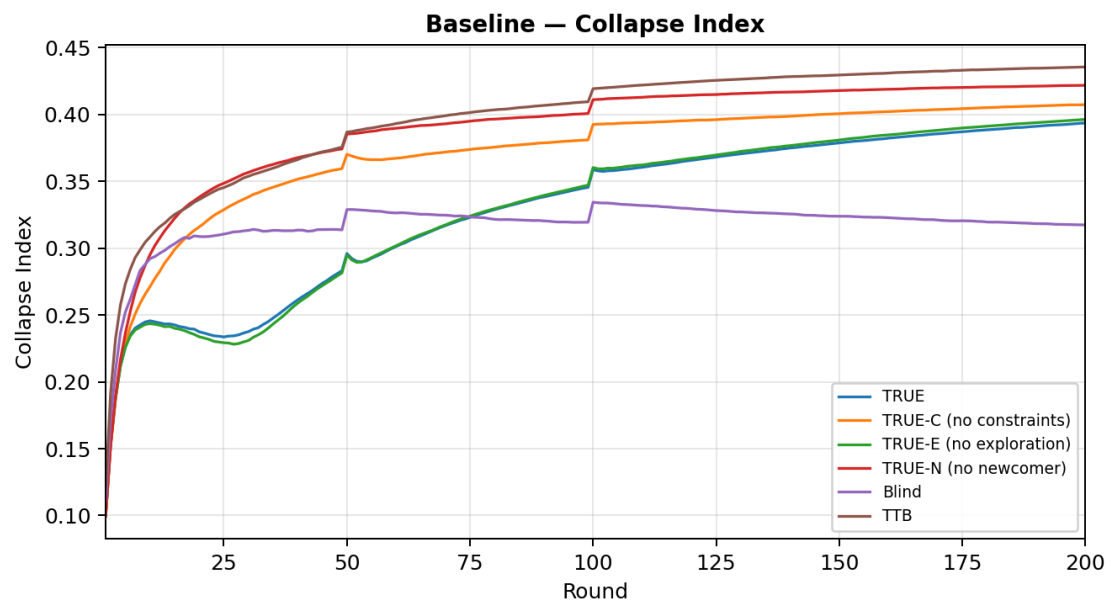
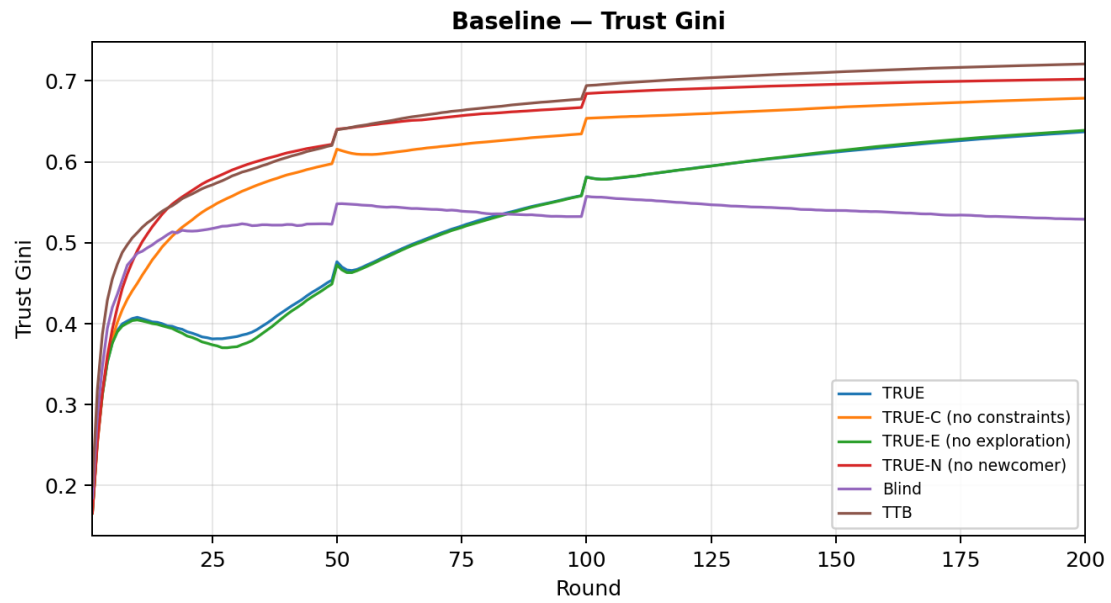
Results Summary

Mean cumulative utility, 95% bootstrap CI, fatal errors, and key metrics per scenario and group.

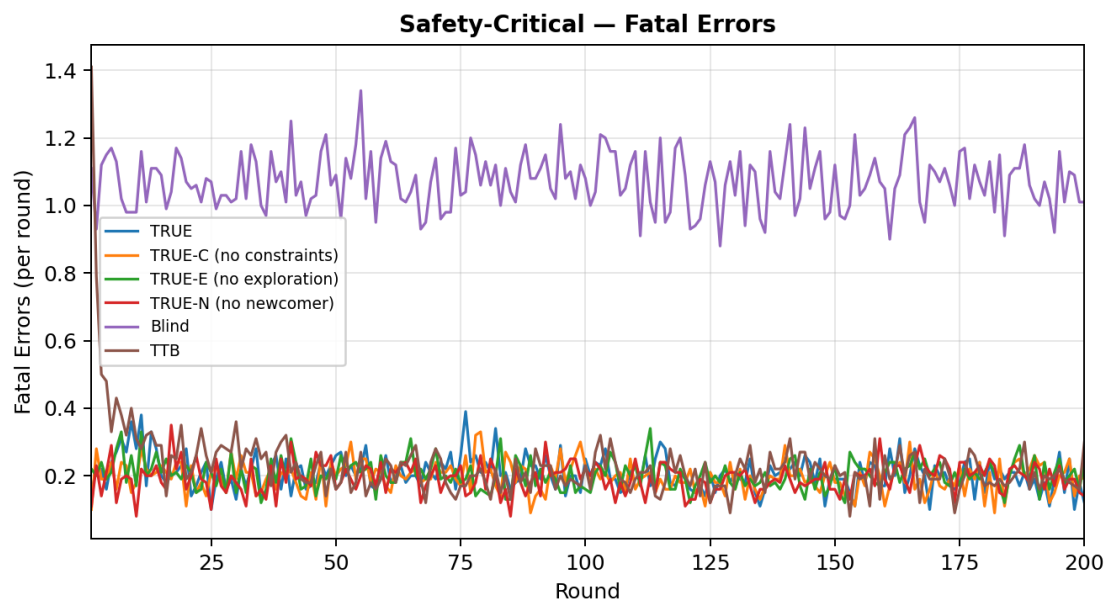
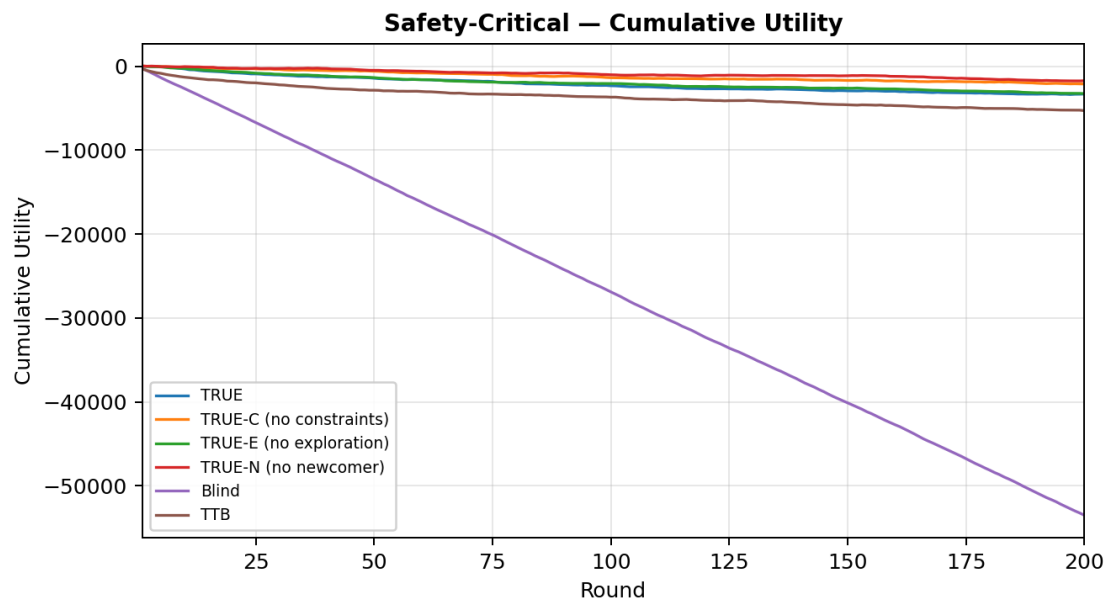
Scenario	Group	Cum.Utility	95% CI	Fatal	True Q	Gini	Collapse	A9 Delay
Baseline	TRUE	896.8	[520.4, 1274.0]	61.0	0.767	0.637	0.394	0.2
Baseline	TRUE-C (no constraints)	1377.6	[1037.2, 1739.0]	60.7	0.799	0.679	0.408	0.9
Baseline	TRUE-E (no exploration)	779.2	[451.0, 1103.0]	61.0	0.765	0.639	0.396	0.2
Baseline	TRUE-N (no newcomer)	1326.0	[974.4, 1671.8]	60.5	0.798	0.702	0.422	383.9
Baseline	Blind	-23753.6	[-24089.4, -23428.6]	208.2	0.343	0.529	0.317	6.8
Baseline	TTB	-2289.2	[-3435.4, -1271.8]	77.2	0.738	0.721	0.436	701.3
Safety-Critical	TRUE	-3313.0	[-3820.2, -2813.8]	41.5	0.794	0.656	0.414	0.4
Safety-Critical	TRUE-C (no constraints)	-2116.4	[-2635.4, -1568.0]	39.3	0.826	0.685	0.418	0.9
Safety-Critical	TRUE-E (no exploration)	-3262.6	[-3825.4, -2707.4]	40.9	0.795	0.655	0.414	0.3
Safety-Critical	TRUE-N (no newcomer)	-1725.8	[-2209.2, -1218.0]	38.8	0.831	0.721	0.438	920.5
Safety-Critical	Blind	-53506.0	[-54045.8, -52926.2]	214.4	0.340	0.538	0.323	7.0
Safety-Critical	TTB	-5295.6	[-6260.4, -4436.8]	46.0	0.775	0.736	0.450	683.1
Observation-Manipulated	TRUE	2102.4	[1752.4, 2443.6]	51.7	0.769	0.602	0.363	0.0
Observation-Manipulated	TRUE-C (no constraints)	2819.6	[2452.4, 3204.4]	50.5	0.805	0.655	0.393	0.5
Observation-Manipulated	TRUE-E (no exploration)	2070.4	[1720.4, 2422.4]	51.7	0.769	0.611	0.368	0.0
Observation-Manipulated	TRUE-N (no newcomer)	2457.2	[2102.8, 2828.4]	51.7	0.802	0.657	0.394	3.8
Observation-Manipulated	Blind	-25168.4	[-25488.0, -24823.6]	198.5	0.335	0.519	0.311	5.0
Observation-Manipulated	TTB	-28398.8	[-29048.8, -27646.0]	221.8	0.233	0.855	0.513	999.0
Utility-Trust-Misalignment	TRUE	2345.4	[2040.2, 2640.5]	56.1	0.749	0.619	0.381	0.1
Utility-Trust-Misalignment	TRUE-C (no constraints)	2954.1	[2640.3, 3284.8]	55.6	0.782	0.659	0.396	0.7
Utility-Trust-Misalignment	TRUE-E (no exploration)	2473.9	[2188.1, 2762.3]	54.9	0.748	0.621	0.384	0.3
Utility-Trust-Misalignment	TRUE-N (no newcomer)	3332.8	[3057.4, 3626.7]	54.2	0.784	0.679	0.408	407.3
Utility-Trust-Misalignment	Blind	-21939.2	[-22229.4, -21643.8]	206.9	0.335	0.533	0.320	7.3
Utility-Trust-Misalignment	TTB	603.8	[-377.7, 1502.8]	65.2	0.731	0.727	0.438	791.1

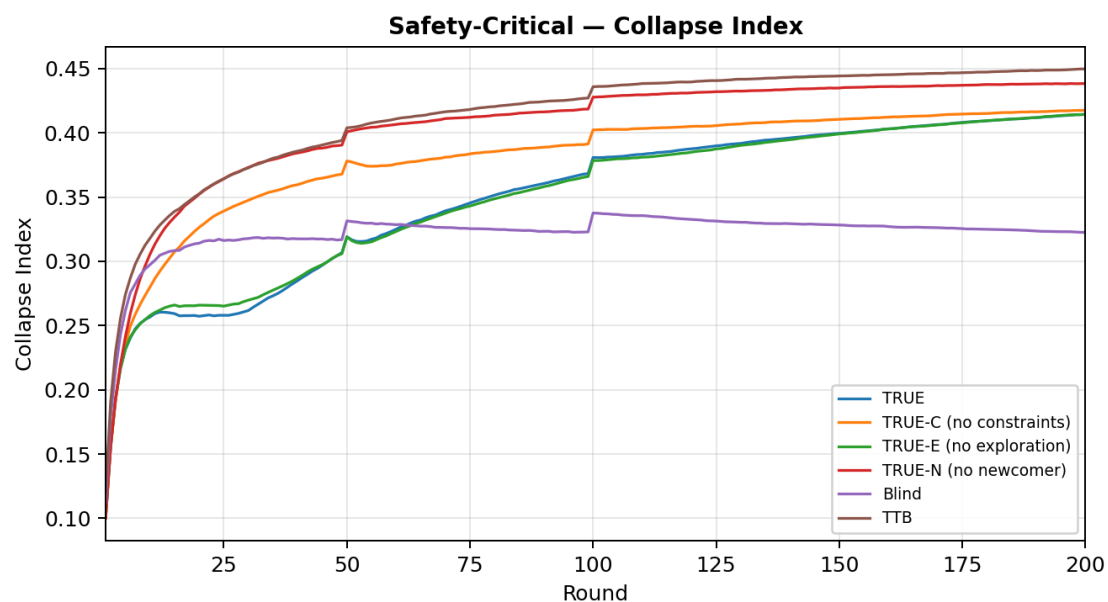
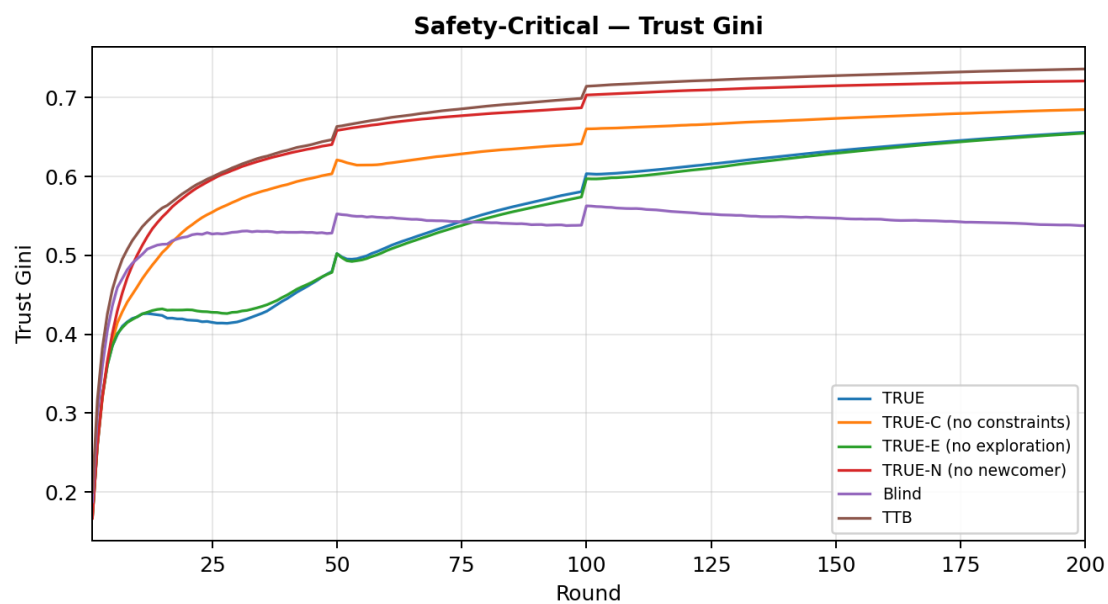
Baseline — Time-Series Analysis



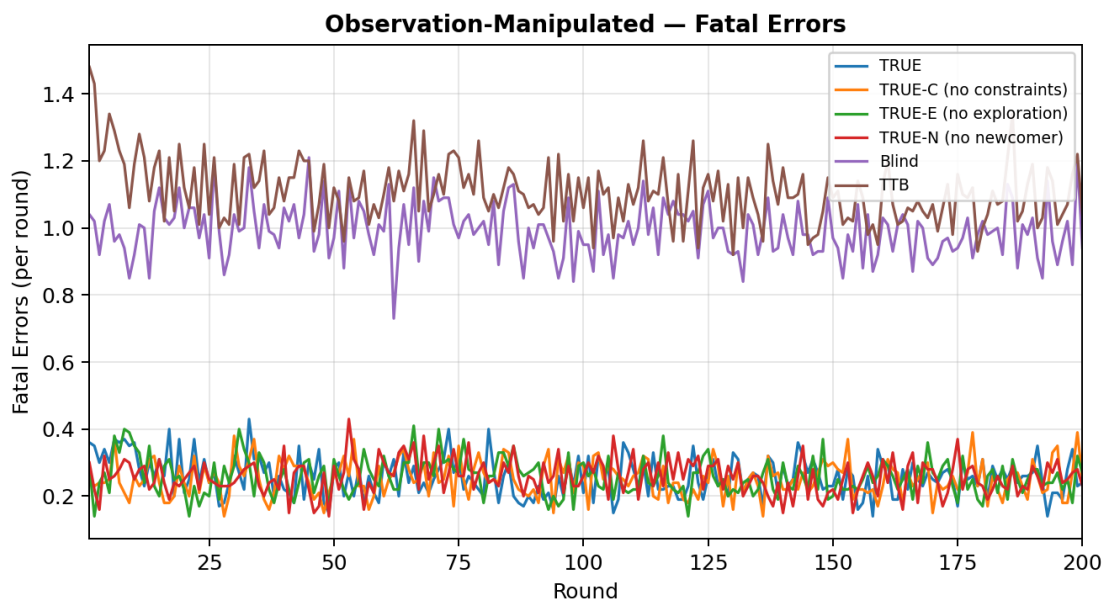
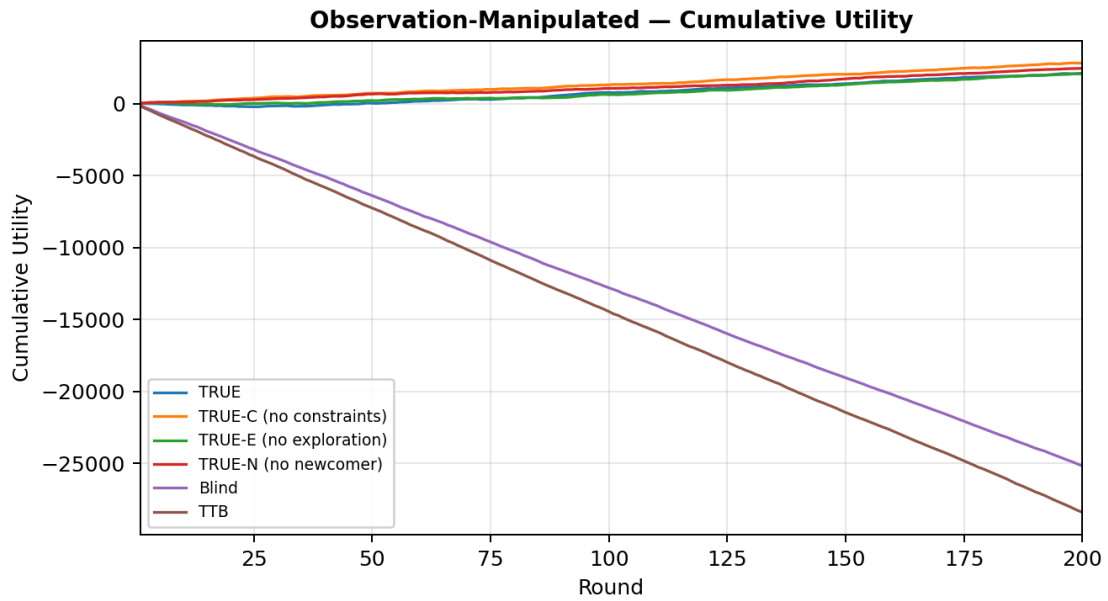


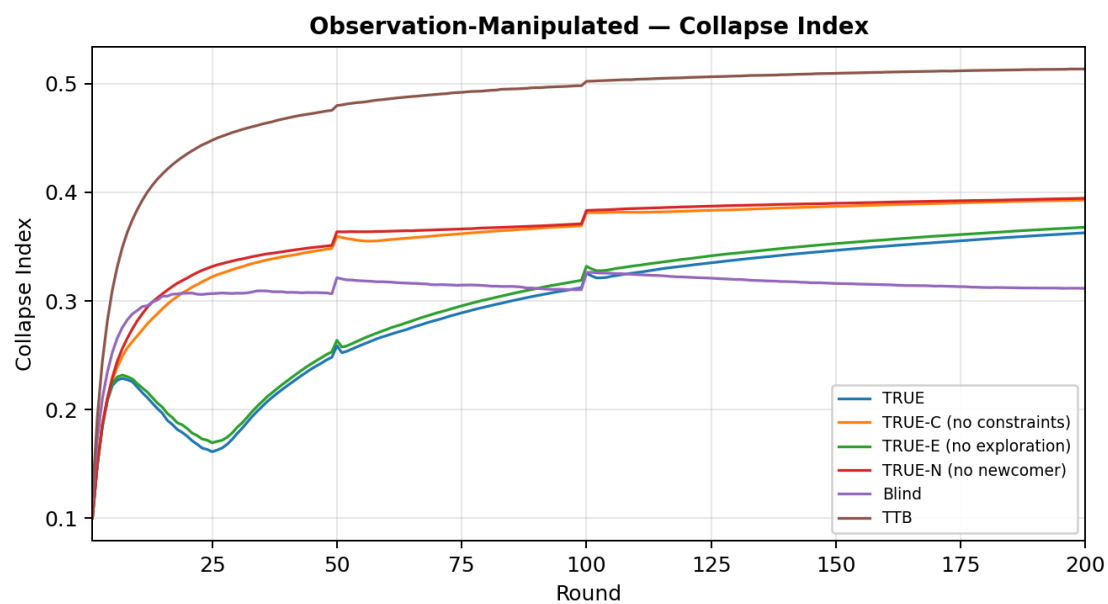
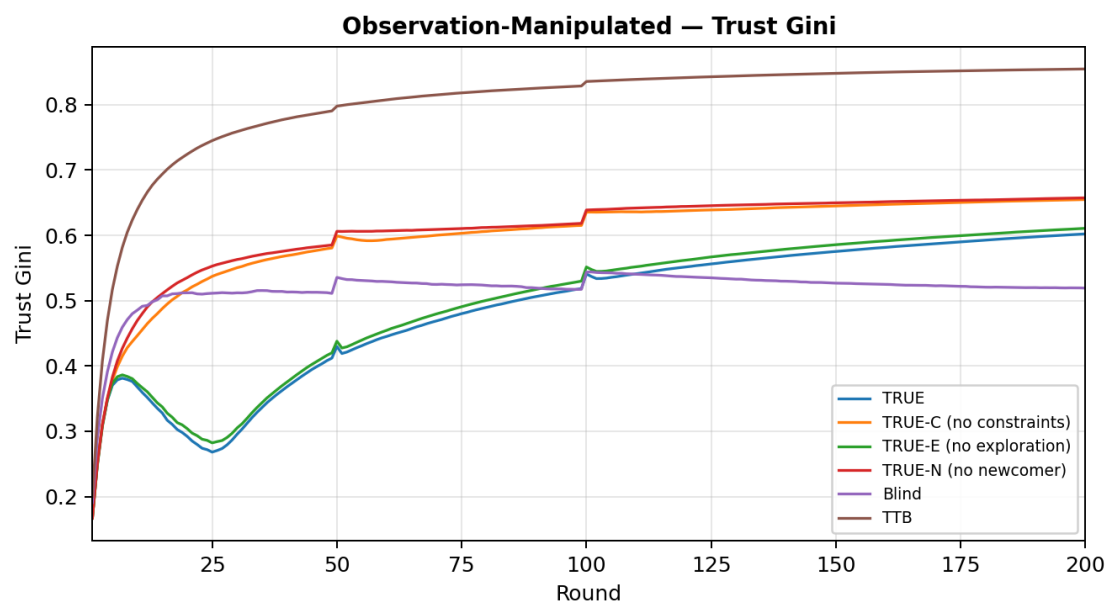
Safety-Critical — Time-Series Analysis



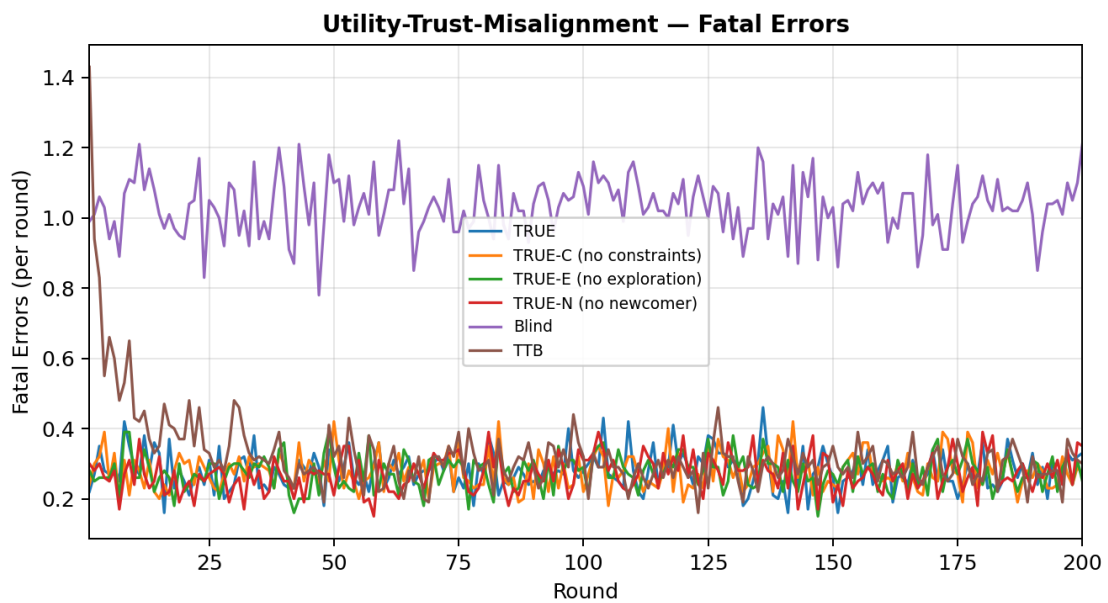
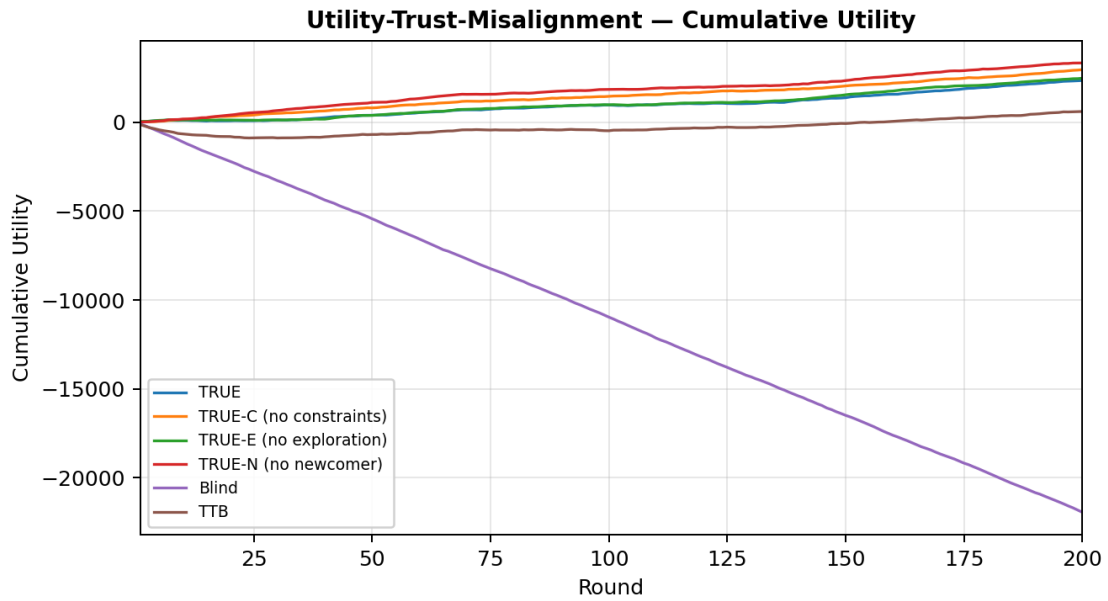


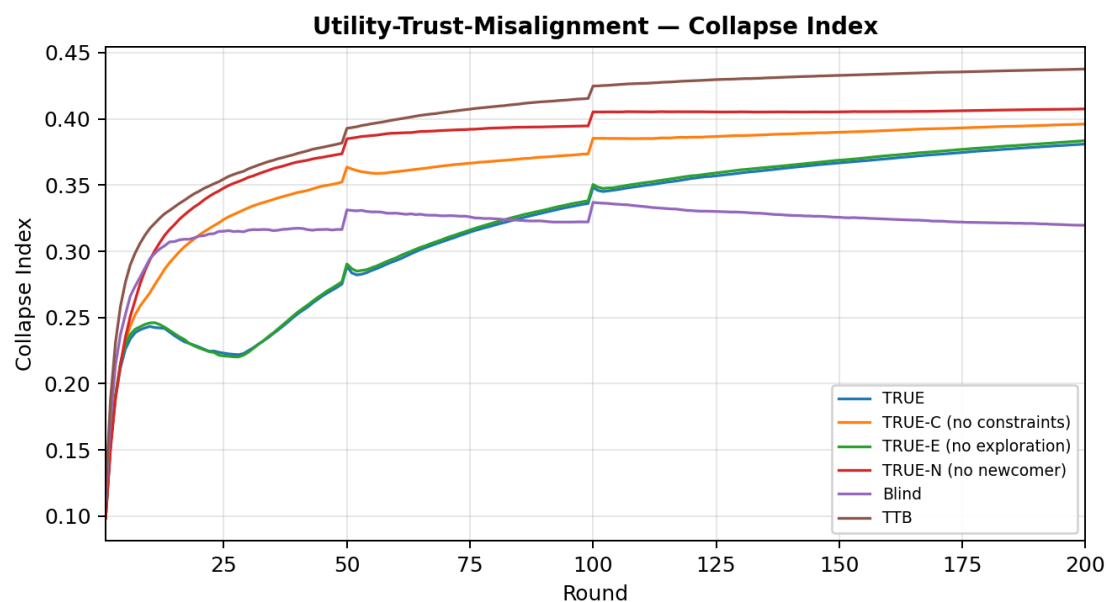
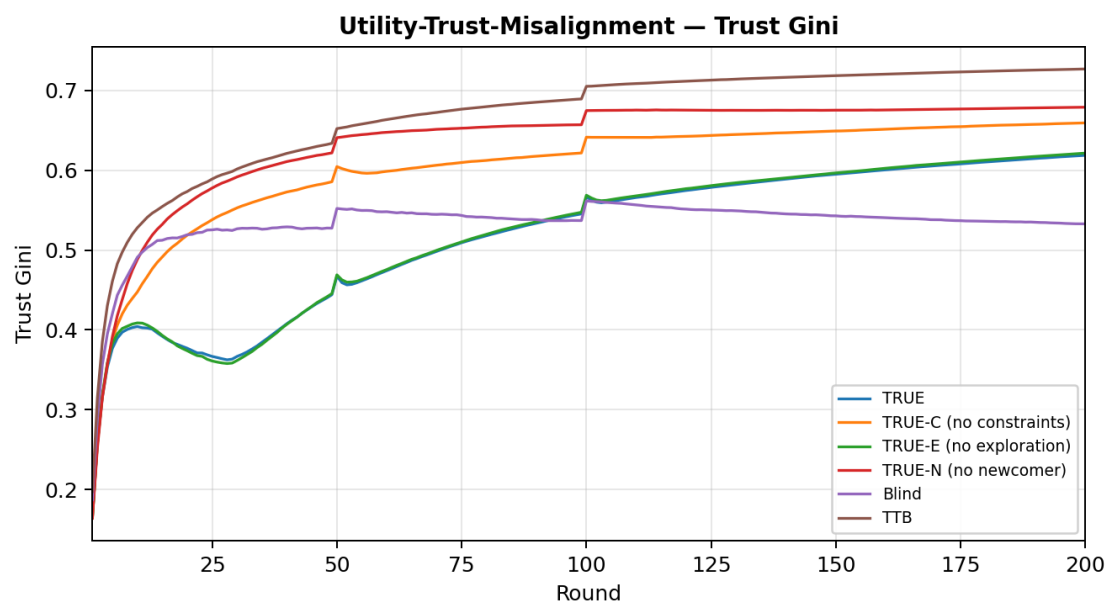
Observation-Manipulated — Time-Series Analysis





Utility-Trust-Misalignment — Time-Series Analysis





Hypothesis Tests (Selected)

All p-values reported with Bonferroni correction across the full test battery.

Scenario	Hypothesis	Mean Diff	t	p_t(Bonf)	Cohen d
Baseline	TRUE_U > Blind_U	24650.4	116.80	0.0000	11.7
Baseline	TRUE_Fatal < Blind_Fatal	147.2	121.14	0.0000	12.1
Baseline	TRUE_U > TTB_U	3186.0	5.32	0.0000	0.5
Baseline	TRUE_Fatal < TTB_Fatal	16.2	5.40	0.0000	0.5
Baseline	TRUE_Collapse < TTB_Collapse	0.0	15.42	0.0000	1.5
Baseline	TRUE_A9 < Blind_A9	6.5	9.15	0.0000	0.9
Baseline	TRUE-C_U > TTB_U	3666.8	6.02	0.0000	0.6
Baseline	TRUE-N_U > TTB_U	3615.2	6.07	0.0000	0.6
Observation-Manipulated	TRUE_U > Blind_U	27270.8	127.77	0.0000	12.8
Observation-Manipulated	TRUE_Fatal < Blind_Fatal	146.8	125.03	0.0000	12.5
Observation-Manipulated	TRUE_U > TTB_U	30501.2	74.13	0.0000	7.4
Observation-Manipulated	TRUE_Fatal < TTB_Fatal	170.0	58.09	0.0000	5.8
Observation-Manipulated	TRUE_Collapse < TTB_Collapse	0.2	118.51	0.0000	11.9
Observation-Manipulated	TRUE_A9 < Blind_A9	5.0	8.62	0.0000	0.9
Observation-Manipulated	TRUE-C_U > TTB_U	31218.4	76.29	0.0000	7.6
Observation-Manipulated	TRUE-N_U > TTB_U	30856.0	77.48	0.0000	7.7
Safety-Critical	TRUE_U > Blind_U	50193.0	133.68	0.0000	13.4
Safety-Critical	TRUE_Fatal < Blind_Fatal	172.9	136.72	0.0000	13.7
Safety-Critical	TRUE_U > TTB_U	1982.6	3.85	0.0114	0.4
Safety-Critical	TRUE_Fatal < TTB_Fatal	4.5	3.34	0.0802	0.3
Safety-Critical	TRUE_Collapse < TTB_Collapse	0.0	11.57	0.0000	1.2
Safety-Critical	TRUE_A9 < Blind_A9	6.6	9.57	0.0000	1.0
Safety-Critical	TRUE-C_U > TTB_U	3179.2	5.90	0.0000	0.6
Safety-Critical	TRUE-N_U > TTB_U	3569.8	6.59	0.0000	0.7
Utility-Trust-Misalignment	TRUE_U > Blind_U	24284.6	118.42	0.0000	11.8
Utility-Trust-Misalignment	TRUE_Fatal < Blind_Fatal	150.8	107.40	0.0000	10.7
Utility-Trust-Misalignment	TRUE_U > TTB_U	1741.6	3.74	0.0178	0.4
Utility-Trust-Misalignment	TRUE_Fatal < TTB_Fatal	9.1	3.85	0.0115	0.4
Utility-Trust-Misalignment	TRUE_Collapse < TTB_Collapse	0.1	25.54	0.0000	2.6
Utility-Trust-Misalignment	TRUE_A9 < Blind_A9	7.2	9.64	0.0000	1.0
Utility-Trust-Misalignment	TRUE-C_U > TTB_U	2350.3	4.66	0.0003	0.5
Utility-Trust-Misalignment	TRUE-N_U > TTB_U	2729.0	5.81	0.0000	0.6

Interpretation & Limitations

Task-flow pairing: Within each run all groups see the same module sequence. Differences are therefore attributable to selection mechanisms, not task luck.

Observation uniformity: A8 surface-quality bonus is uniform across groups. The only A8 asymmetry is inside TTB's scoring function (A8 receives extra trust-objective weight), which is a mechanism-level difference.

Ablation surprise: TRUE-C and TRUE-N outperform full TRUE in cumulative utility under current parameters, suggesting that constraint filtering and newcomer protection may be overly conservative. However, TRUE-N shows dramatically higher A9 cold-start delay, confirming that newcomer protection serves its intended purpose.

Bonferroni correction: Conservative; hypotheses remaining significant after correction are robust.